

datefunctions

March 1, 2026

PySpark Date Functions

```
python from pyspark.sql.functions import *
```

Current Date & Time

Function	Syntax	Description
<code>current_date()</code>	<code>df.withColumn("today", current_date())</code>	Returns current system date
<code>current_timestamp()</code>	<code>df.withColumn("now", current_timestamp())</code>	Returns current timestamp
<code>localtimestamp()</code>	<code>df.withColumn("lt", localtimestamp())</code>	Session local timestamp
<code>now()</code>	<code>df.withColumn("now", now())</code>	Same as current_timestamp

Date Formatting & Conversion

Function	Syntax	Description
<code>to_date()</code>	<code>df.withColumn("dt", to_date("date_str", "yyyy-MM-dd"))</code>	String → Date
<code>to_timestamp()</code>	<code>df.withColumn("ts", to_timestamp("date_str", "yyyy-MM-dd HH:mm:ss"))</code>	String → Timestamp
<code>date_format()</code>	<code>df.withColumn("fmt", date_format("dt", "dd-MM-yyyy"))</code>	Format date
<code>unix_timestamp()</code>	<code>df.withColumn("unix", unix_timestamp("date_str", "yyyy-MM-dd"))</code>	Converts to Unix time
<code>from_unixtime()</code>	<code>df.withColumn("date", from_unixtime("unix"))</code>	Unix → Timestamp
<code>cast("date")</code>	<code>df.withColumn("dt", col("date_str").cast("date"))</code>	Type casting
<code>cast("timestamp")</code>	<code>df.withColumn("ts", col("date_str").cast("timestamp"))</code>	Type casting

Extract Date Parts

Function	Syntax	Description
<code>year()</code>	<code>df.withColumn("year", year("dt"))</code>	Extract year
<code>month()</code>	<code>df.withColumn("month", month("dt"))</code>	Extract month
<code>dayofmonth()</code>	<code>df.withColumn("day", dayofmonth("dt"))</code>	Day of month
<code>dayofyear()</code>	<code>df.withColumn("doy", dayofyear("dt"))</code>	Day of year
<code>weekofyear()</code>	<code>df.withColumn("week", weekofyear("dt"))</code>	Week number
<code>quarter()</code>	<code>df.withColumn("qtr", quarter("dt"))</code>	Quarter
<code>hour()</code>	<code>df.withColumn("hr", hour("ts"))</code>	Extract hour
<code>minute()</code>	<code>df.withColumn("min", minute("ts"))</code>	Extract minute
<code>second()</code>	<code>df.withColumn("sec", second("ts"))</code>	Extract second

Date Arithmetic

Function	Syntax	Description
<code>date_add()</code>	<code>df.withColumn("new", date_add("dt",10))</code>	Add days
<code>date_sub()</code>	<code>df.withColumn("new", date_sub("dt",10))</code>	Subtract days
<code>datediff()</code>	<code>df.withColumn("diff", datediff("end","start"))</code>	Difference in days
<code>add_months()</code>	<code>df.withColumn("new", add_months("dt",2))</code>	Add months
<code>months_between()</code>	<code>df.withColumn("mb", months_between("end","start"))</code>	Months difference
<code>next_day()</code>	<code>df.withColumn("next", next_day("dt","Sunday"))</code>	Next specific weekday
<code>last_day()</code>	<code>df.withColumn("last", last_day("dt"))</code>	Last day of month

Date Truncation & Rounding

Function	Syntax	Description
<code>trunc()</code>	<code>df.withColumn("month_start", trunc("dt","month"))</code>	Truncate to month
<code>date_trunc()</code>	<code>df.withColumn("hour_trunc", date_trunc("hour","ts"))</code>	Truncate timestamp
<code>floor()</code>	<code>df.select(floor(months_between("end","start")))</code>	Floor value
<code>ceil()</code>	<code>df.select(ceil(months_between("end","start")))</code>	Ceil value

Timestamp & Timezone

Function	Syntax	Description
<code>from_utc_timestamp()</code>	<code>df.withColumn("ist", from_utc_timestamp("ts","Asia/Kolkata"))</code>	Convert UTC → Timezone
<code>to_utc_timestamp()</code>	<code>df.withColumn("utc", to_utc_timestamp("ts","Asia/Kolkata"))</code>	Convert local → UTC

Common Exam Patterns (Very Important)

Use Case	Syntax
Calculate Age	<code>floor(months_between(current_date(),"dob")/12)</code>
Last 7 days data	<code>df.filter(col("dt") >= date_sub(current_date(),7))</code>
Month start	<code>trunc("dt","month")</code>
Month end	<code>last_day("dt")</code>
Difference between two timestamps	<code>datediff(to_date("ts1"), to_date("ts2"))</code>
Convert string to proper date	<code>to_date("col","yyyy-MM-dd")</code>

Important Format Patterns

Pattern	Meaning
yyyy	4-digit year
yy	2-digit year
MM	Month (01–12)
dd	Day
HH	Hour (24)
mm	Minute
ss	Second

<https://chatgpt.com/share/69a3df3b-a858-8002-8a26-9a3e83d8674b>